MA341
Applied ODE's
Homework 3.5

1. Use Euler's Method for Linear Systems of ODE to find **real valued** set of fundamental solutions

   (a) $\frac{d}{dt}X = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} X$ (b) $\frac{d}{dt}X = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} X$ (c) $\frac{d}{dt}X = \begin{pmatrix} 4 & 3 \\ -6 & -2 \end{pmatrix} X$

2. Use Euler's Method for Linear Systems of ODE to solve the following initial value problems.

   (a) $\frac{d}{dt}X = \begin{pmatrix} 1 & -1 \\ 5 & -3 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$

   (b) $\frac{d}{dt}X = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

   (c) $\frac{d}{dt}X = \begin{pmatrix} 4 & 0 & -3 \\ -3 & 1 & 3 \\ 6 & 0 & -5 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$

3. Consider $y'''(t) - y''(t) - y'(t) + y(t) = 0$

   (a) Denote new variables $x_1(t) := y(t), \quad x_2(t) := y'(t), \quad x_3(t) := y''(t)$ and solve the following system

   $$X'(t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} X(t), \quad X(t) := \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \begin{pmatrix} y(t) \\ y'(t) \\ y''(t) \end{pmatrix}$$

   (b) Use your solution to the system to find the solution to the original equation (verify!).

4. Consider the following initial value problem:

   $$t\mathbf{x}'(t) = \begin{bmatrix} 3 & -2 \\ 2 & -2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} -2t \\ t^4 - 1 \end{bmatrix}, \mathbf{x}(1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

   (a) Show that $\mathbf{x}_1(t) = \begin{bmatrix} t^{-1} \\ 2t^{-1} \end{bmatrix}$ and $\mathbf{x}_2(t) = \begin{bmatrix} 2t^2 \\ t^2 \end{bmatrix}$ are the fundamental solutions to the associated homogeneous system.

   (b) Use variation of parameters for systems to find the solution to the problem.

5. Solve the following problems. **Note**: In those problems the geometric multiplicity is less than algebraic multiplicity.

   (a) $\frac{d}{dt}X = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix} X \quad X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$

   (b) $\frac{d}{dt}X = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 3 & 6 & 2 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} -1 \\ 2 \\ -30 \end{pmatrix}$